American Computer Science League

2025-2026 ● Contest 4: Numbrix ● Junior Division

PROBLEM: *Numbrix* is an *n* x *n* grid number puzzle. The objective is to fill the grid with all of the integers from 1 to n^2 inclusive by placing all missing integers, *x*, in the grid such that both *x - 1* and *x + 1* are adjacent to it, in the same row or the same column.

In this problem, you will be given the grid size, *n*, and the integers 1 through n^2 that fill the grid in row-major order. All missing numbers are represented by 0. Your task is to determine the missing numbers and output them in row-major order.

We guarantee that:

- Missing numbers will never be adjacent to each other, horizontally, vertically, or diagonally.
- The numbers 1 and n^2 will never be one of the missing numbers.
- There will be no more than $2n$ missing numbers.

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EXAMPLE:

<table>
<tr><td>Input</td><td>4
1 4 5 0 0 3 8 7 11 10 0 16 0 13 14 15</td></tr>
<tr><td>Output</td><td>6 2 9 12</td></tr>
<tr><td>Explanation</td><td>The puzzle is represented in the 2-dimensional array as follows:
<table><tr><td>1</td><td>4</td><td>5</td><td>0</td></tr><tr><td>0</td><td>3</td><td>8</td><td>7</td></tr><tr><td>11</td><td>10</td><td>0</td><td>16</td></tr><tr><td>0</td><td>13</td><td>14</td><td>15</td></tr></table>The missing integers from the grid are 2, 6, 9, and 12. They are placed as follows:
<table><tr><td>1</td><td>4</td><td>5</td><td>6</td></tr><tr><td>2</td><td>3</td><td>8</td><td>7</td></tr><tr><td>11</td><td>10</td><td>9</td><td>16</td></tr><tr><td>12</td><td>13</td><td>14</td><td>15</td></tr></table>Output those integers in row-major order: 6 2 9 12.</td></tr>
</table>

INPUT: Input an integer for the size of the puzzle, n, and a string containing the n^2 integers that fill the grid in row-major order, with missing numbers represented by a 0. The value of n will be between 4 and 9, inclusive.

OUTPUT: Output a string of integers representing the missing number(s), in row-major order, each separated by a single space.

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SAMPLE INPUT:

1. 4
   1 4 5 0 0 3 8 7 11 10 0 16 0 13 14 15
2. 4
   16 5 6 7 0 4 0 8 14 1 2 9 0 12 0 10
3. 5
   13 14 15 16 0 12 0 6 5 18 11 8 1 4 0 10 9 0 3 20 25 24 23 22 0
4. 6
   16 15 4 5 0 7 0 14 0 2 1 8 18 13 12 11 10 0 19 26 27 0 31 32 0 25 24 29 30 0 21 22 0 36 35 34
5. 5
   21 0 5 6 7 20 23 4 3 8 19 0 25 0 9 18 15 14 1 10 0 16 0 12 0
6. 6
   1 0 5 8 9 10 2 3 6 0 12 11 27 26 21 20 13 14 28 0 22 19 16 0 29 24 23 18 17 36 0 31 32 0 34 35

SAMPLE OUTPUT:

1. 6 2 9 12
2. 15 3 13 11
3. 17 7 19 2 21
4. 6 17 3 9 28 20 33 23
5. 22 24 2 17 13 11
6. 4 7 25 15 30 33

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TEST DATA

TEST INPUT:

1. 4
   2 3 4 0 1 0 7 6 10 9 16 15 0 12 0 14
2. 5
   17 16 15 0 11 18 1 14 13 10 19 2 0 6 9 0 3 4 7 8 21 22 23 0 25
3. 6
   5 0 7 0 13 14 4 3 8 11 16 0 1 2 9 0 17 18 32 0 28 27 26 19 33 30 29 24 0 20 34 0 36 23 22 21
4. 7
   39 38 37 8 0 4 3 0 41 36 9 6 5 0 43 42 0 10 13 14 1 44 33 34 11 12 0 16 45 0 29 0 27 26 17 46 31 30 23 24 25 0 47 0 49 0 21 20 19
5. 7
   3 2 1 42 43 44 49 4 0 6 41 0 45 48 9 8 7 38 39 46 0 10 0 18 0 36 35 34 11 16 19 20 21 32 33 0 15 24 23 22 31 0 13 14 0 26 0 28 29
6. 8
   3 0 33 34 37 38 41 42 4 1 32 35 36 39 40 43 5 6 31 24 23 46 45 0 8 7 30 0 22 0 52 53 9 0 29 26 21 48 51 0 12 11 28 27 20 49 50 55 13 0 17 0 19 60 0 56 14 15 64 63 62 61 58 57

TEST OUTPUT:

1. 5 8 11 13
2. 12 5 20 24
3. 6 12 15 10 31 25 35
4. 7 40 2 35 15 32 28 18 48 22
5. 5 40 47 17 37 12 30 25 27
6. 2 44 25 47 10 54 16 18 59

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问题: *Numbrix* 是一个 n x n 的网格数字谜题。目标是将从 1 到 n^2 的所有整数填入网格，即把所有缺失的整数 x 放入网格中，使得 $x - 1$ 和 $x + 1$ 分别与 x 在同一行或同一列相邻。

在本题中，已知网格的大小 n，以及按行优先顺序填充网格的整数 1 到 n^2。所有缺失的数字用 0 表示。你的任务是确定缺失的数字，并按行优先顺序输出它们。

我们需确保：

- 缺失的数字之间永远不会在水平、垂直或对角方向上相邻。
- 数字 1 和 n^2 永远不会是缺失的数字。
- 缺失的数字个数不会超过 $2n$ 。

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示例:

输入	4 1 4 5 0 0 3 8 7 11 10 0 16 0 13 14 15
输出	6 2 9 12
解析	该谜题在二维数组中的表示方式如下： 网格中缺失的整数是 2、6、9 和 12。它们的位置如下所示： 按行优先顺序输出这些整数: 6 2 9 12.

1	4	5	0
0	3	8	7
11	10	0	16
0	13	14	15

1	4	5	**6**
2	3	8	7
11	10	**9**	16
12	13	14	15

输入: 输入一个整数表示网格的尺寸 n，以及一个字符串，该字符串中包含按行优先顺序填充网格的 n^2 个整数，缺失的数字用 0 表示。n 的取值范围是 4 到 9（包含两端）。
输出: 输出一个由整数组成的字符串，表示按行优先顺序排列的缺失数字，每个数字之间用单个空格分隔。

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样例输入：

1. 4
   1 4 5 0 0 3 8 7 11 10 0 16 0 13 14 15
2. 4
   16 5 6 7 0 4 0 8 14 1 2 9 0 12 0 10
3. 5
   13 14 15 16 0 12 0 6 5 18 11 8 1 4 0 10 9 0 3 20 25 24 23 22 0
4. 6
   16 15 4 5 0 7 0 14 0 2 1 8 18 13 12 11 10 0 19 26 27 0 31 32 0 25 24 29 30 0 21 22 0 36 35 34
5. 5
   21 0 5 6 7 20 23 4 3 8 19 0 25 0 9 18 15 14 1 10 0 16 0 12 0
6. 6
   1 0 5 8 9 10 2 3 6 0 12 11 27 26 21 20 13 14 28 0 22 19 16 0 29 24 23 18 17 36 0 31 32 0 34 35

样例输出：

1. 6 2 9 12
2. 15 3 13 11
3. 17 7 19 2 21
4. 6 17 3 9 28 20 33 23
5. 22 24 2 17 13 11
6. 4 7 25 15 30 33

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测试数据

测试输入：

1. 4
   2 3 4 0 1 0 7 6 10 9 16 15 0 12 0 14
2. 5
   17 16 15 0 11 18 1 14 13 10 19 2 0 6 9 0 3 4 7 8 21 22 23 0 25
3. 6
   5 0 7 0 13 14 4 3 8 11 16 0 1 2 9 0 17 18 32 0 28 27 26 19 33 30 29 24 0 20 34 0 36 23 22 21
4. 7
   39 38 37 8 0 4 3 0 41 36 9 6 5 0 43 42 0 10 13 14 1 44 33 34 11 12 0 16 45 0 29 0 27 26 17 46 31 30 23 24 25 0 47 0 49 0 21 20 19
5. 7
   3 2 1 42 43 44 49 4 0 6 41 0 45 48 9 8 7 38 39 46 0 10 0 18 0 36 35 34 11 16 19 20 21 32 33 0 15 24 23 22 31 0 13 14 0 26 0 28 29
6. 8
   3 0 33 34 37 38 41 42 4 1 32 35 36 39 40 43 5 6 31 24 23 46 45 0 8 7 30 0 22 0 52 53 9 0 29 26 21 48 51 0 12 11 28 27 20 49 50 55 13 0 17 0 19 60 0 56 14 15 64 63 62 61 58 57

测试输出：

1. 5 8 11 13
2. 12 5 20 24
3. 6 12 15 10 31 25 35
4. 7 40 2 35 15 32 28 18 48 22
5. 5 40 47 17 37 12 30 25 27
6. 2 44 25 47 10 54 16 18 59